

GRID CURRENT THD REDUCTION IN TWO-STAGE GRID-TIED PV-MICROINVERTER USING PASSIVE DAMPING PD-FILTER

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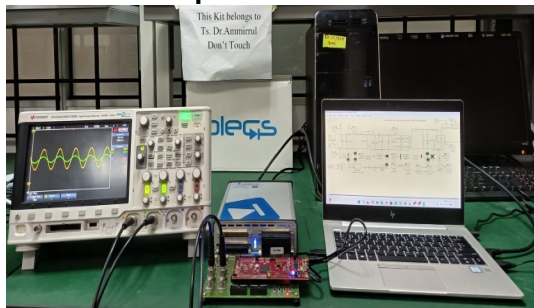
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Graphical abstract



Abstract

This paper presents analysis of two-stage grid-tied Photovoltaic (PV) microinverter, confirming its designing, modelling and performance improvements reducing Total Harmonic Distortion (THD) by IEEE power quality standards. The study evaluates three various filter configurations integrated into microinverter, each designed to mitigate THD in the grid current, the first configuration implement LCL filter, obtaining THD of approximately 11%, the second configuration introduces hybrid filtering approach that combines the LCL filter with hybrid Half Bridge (H-B) filter, demonstrating improvements and yielding THD values between 6-7%, this combination efficient addresses high frequency noise, enhancing output quality, the third configuration utilizes hybrid LCL and Passive Damping (PD) filter, this developed arrangement results in the lowest THD levels, reaching as low as 3-4%, the passive damping component stabilizes the system by mitigating resonances and improving performance at varying operation conditions, Experimental results reflect steady-state operation verification through Hardware-in-the-Loop (HIL) testing using (PLECS), with measurements captured via KEYSIGHT oscilloscope, these results exhibit full compatibility with simulation outcomes obtained from MATLAB/Simulink, confirming the efficacy of the proposed filtering strategies in obtaining reduced harmonic distortion, this research design and optimization of efficient grid-compliant PV-microinverter systems, addressing power quality concerns enhances grid stability.

Keywords: Two-stage grid-tied PV-microinverter, Passive Damping (PD) filter, Total Harmonic Distortion (THD).

1.0 INTRODUCTION

Traffic management and accident management is very important for human in congestion city in order to provide

comfortable lifestyle and safety. By having this system, people will have real-time information on traffic condition around the city, which can avoid from stuck on traffic jam. At present, ground-based solution are widely used to monitor

traffic. Energy is main component of modern society, driving economic growth, technological advancement, and quality of life, in 2024 global energy landscape increasing influenced by urgency of climate change and the need for sustainable development, this transition has accelerated deployment of renewable energy technologies as alternatives to fossil-based generation sources [1]. Within solar energy applications, PV-microinverters represent advance in distributed generation architectures, these devices convert DC current, solar modules output, into grid-synchronized AC current, microinverters operate at module level, enabling decentralized power condition and independent energy extraction [2].

Grid-tied PV microinverter system provide module-level maximum power optimization, ensuring that each panel operates efficiently non-uniform environmental conditions such as partial shading, or temperature variations, this decentralized configuration improves overall energy yield, simplifies installation logistics, and supports adaptable system layouts suitable for diverse structural designs [3]. Their modular structure facilitates incremental system expansion without architectural modifications, reducing operational costs and enhancing economic feasibility [4]. Grid-tied PV-microinverters enhance solar energy systems by maximizing energy generation, ensuring safety, and simplifying management, making them excellent options for increasing renewable energy [5].

Two-stage single-phase microinverter topology, voltage conditioning is achieved through dedicated DC–DC conversion stage, the quadratic boost converter is adopted due to its ability to attain high voltage gain while maintaining moderate component requirements, quadratic boost converter enhances voltage amplification capability and broadens permissible input voltage range, this contributes to improved conversion efficiency and stable interfacing with grid voltage levels [6].

Two-stage microinverter design, first stage implements quadratic boost converter to enhance input voltage, this converter features topology with additional capacitors and inductors, allowing it to obtain higher voltage gain with fewer components [7]. Provides flexible operation via wide range of input voltages, enhance microinverter system's reliability, the efficient voltage boost also facilitates better suitable inverter output to grid voltage, improves integration and compliance with grid standards [8].

H-bridge inverters require careful management of switching losses and advanced control strategies to ensure high efficiency and minimal distortion, despite these technical considerations, this topology remains fundamental in single-phase power conversion systems [9]. LCL filter allows fundamental components to pass while suppressing switching-induced distortions, its multi-element structure introduces potential resonance that must be addressed through new design [10]. Operation of LCL filter is focused on attenuating high frequency harmonics to maintain cleaner sine wave output, the combination of these components provides controlled impedance to harmonics, helping stabilize grid conditions and mitigate resonance issues in the system [11].

Compact design balances performance at space requirements, and ability to dampen resonances enhances

system reliability, and control complexities arising from interactions between filter components [12]. These filters are implemented in various applications, including grid-tied inverters for photovoltaic systems [13]. LCL filters are components in modern power electronics, providing to cleaner power output, improved system performance, and compliance with power quality regulations [14]. Passive damping filter introduces minor resistive losses, it offers structural simplicity, robustness, and cost-effectiveness, hybrid configurations combining LCL filter and passive damping filter improved stability and harmonic performance in grid-tied PV microinverter systems [15]. LCL filter is implemented to filter the output of inverters and reduce high-frequency harmonics [16]. Advantages of passive damping filters include their simplicity, cost-effectiveness, and stability, they produce energy losses due to resistive heating, affecting system efficiency [17]. Hybrid filter include passive damping filters improves power quality in two-stage grid-connected PV-microinverters, this configuration enhances system stability and optimizes system performance [18].

Proposed two-stage PV-microinverter, achieving optimal performance necessitates precise filter parameter selection, along with coordinated control implementation [19]. The output from this stage feeds into the inverter in second stage, converting boosted DC voltage into AC voltage are suitable for grid systems [20]. Hybrid filter ensures that generated AC meets grid standards, the advantages include improve harmonic attenuation, enhanced stability due to damping resistor, compact design allowing smaller component sizes, and flexibility to conform to varying grid conditions and loads [21].

Considerations for effective design include exactly sizing the components (L, C, and R) to obtain required performance and implementing an effective control strategy to manage inverter and filter operation [22]. Hybrid filter enhances the performance by controlling harmonic distortion and improving system stability [23]. Advanced design and optimization of the component values are essential for efficient and reliable operation in grid-tied environment [24].

This paper explains topic one the introduction of grid-tied photovoltaic system, that features two-stage PV-microinverter, integrating quadratic boost converter and hybrid filter comprised of LCL filter and PD filter. In topic two, titled two-stage PV-microinverter and damping filter topology, discussion commences with theoretical framework for developed system, outlining its principles and operational features. Topic three of the paper is experimental results of grid current for three different filter topologies, which are presented and discussed, the analysis is conducted at steady state conditions, Hardware-in-the-Loop HIL experiment results using PLECS are explained.

2.0 METHODOLOGY

Two-stage grid-tied PV-microinverter diagram shows in Fig. 1, explains energy flow from PV array to grid system, represents various components that enhance system efficiency and power quality.

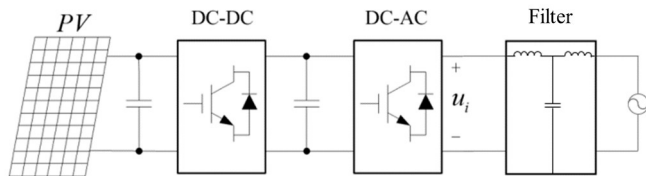


Figure 1 Two-stage PV-microinverter block diagram

First, PV captures sunlight and converts it into direct electricity, that works as power source for PV microinverter, next, DC output from solar array is sent to quadratic boost converter, which steps up the low voltage generated by solar panels to higher voltage level suitable for the inverter stage [25].

Boosted DC output is directed to DC-link capacitor following boost converter, this capacitor smooths output voltage and provides energy storage and stabilize the voltage level.

Next component is H-bridge inverter, that converts boosted DC voltage into AC voltage suitable for grid connection, H-bridge inverter allows for flexible control of output waveform, enabling accurate modulation of AC voltage to meets grid requirements [26].

AC output passes through LCL filter, this filter reduces high-frequency harmonics generated by the inverter, in parallel with LCL filter is PD filter, which includes resistor connected across the capacitor of LC filter, the PD filter dampens resonances and stabilizes system performance by mitigation potential oscillations that may arise due to reactive components. Finally, AC output voltage fed into grid system, allowing PV-microinverter to produce high-power quality to utility grid, enhancing efficient power distribution and utilization [27].

PV-microinverter reduces THD in grid current through various mechanisms [28], first, by implementing LCL filter

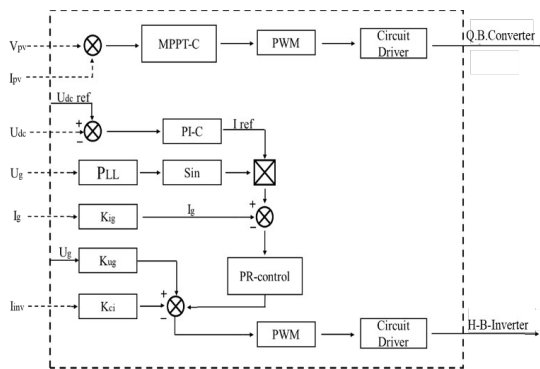


Figure 3 PV-microinverter control scheme

Maximum power point tracking, at main of control strategy, is MPPT algorithm, that constant evaluate operating conditions of solar array to increase maximum

between H-bridge inverter and grid system, effective system attenuates high-frequency harmonics generated during the inversion process, this filtering ensures that AC waveform similar as pure sine wave and have lowering THD, PD filter parallel connected to LCL filter helps system stabilize and mitigate resonances, that may lead to increased harmonic distortion, PD filter dampens oscillations, enhancing output current quality [29].

Schematic of two-stage PV-microinverter shows in Fig. 2. cutting edge design to optimize solar energy conversion from PV array to grid system, this setup is design to enhance energy efficiency and minimize grid current THD, that is important for maintain grid stability and compliance with IEEE and IEC standards.

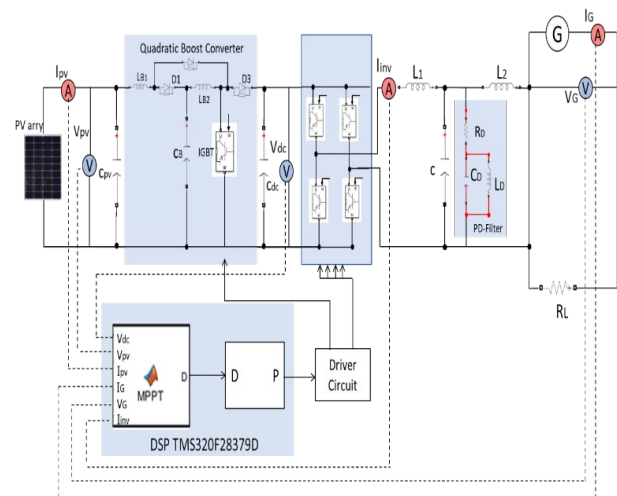


Figure 2 System prototype architecture of PV

The control system combines various necessary components, maximum power point tracking (MPPT) algorithm, proportional resonant (PR) regulator, and proportional integral (PI) controller, each play important role in ensuring optimal operation, control scheme diagram of PV-microinverter shows in Fig. 3.

possible output power, solar irradiance and temperature conditions changes throughout the day, leading to varying output features of solar panels.

MPPT controller implemented an algorithm as perturb and observe (P&O), adjusts duty cycle of quadratic boost converter based on real-time measurement of PV voltage and current, proportional resonant PR regulator, once the energy is conditioned and converted to AC by H-bridge inverter, PR regulator comes to effect, this controller is especially implemented for grid-tied applications, as it can efficiently regulate output current and voltage to meets grid conditions [30].

3.0 MODELLING OF PD HYBRID FILTER

LCL filter consists of following components.

- Input Voltage. V_{in} DC input from PV panels.

- Inverter. converts DC to AC and generates output voltage.
- Inductors L_1 and L_2 . used to limit current change rate and filtering high frequency harmonics.
- Capacitor C . provides low impedance path for high frequency currents and filtering the output.
- Resistance load R_{Load} . shows grid or load connected to grid system.

LCL filter is third-order system, its behavior analyzed using Kirchhoff's laws and Laplace transforms.

Mathematical modelling of LCL filter, impedance of LCL filter in frequency domain s-domain gives by.

$$Z(s) = R_{Load} + sL_2 + sC + sL_1 \quad (1)$$

The equation explains impedance of LCL filter in frequency domain s-domain.

- R_{Load} load resistance, explaining grid or load connected to grid system.
- sL_1 a sL_2 Impedances of L_1 and L_2 , inductors oppose changes in current, and their impedance increases at frequency $s = j\omega$.
- sC Impedance of the capacitor C , capacitor opposes changes in voltage, and their impedance decreases at frequency.
- LCL filter is third-order system because it includes two inductors and one capacitor.

Transfer Function of LCL filter.

- LCL filter transfer function $H(s)$ is ratio of output voltage $V_{out}(s)$ to input voltage $V_{in}(s)$.

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{Z(s)} \quad (2)$$

- LCL filter's transfer function is ratio of output voltage $V_{out}(s)$ to input voltage $V_{in}(s)$.
- Transfer function describes LCL filter amplifies different frequencies in input signal.
- $Z(s)$ LCL filter total impedance, $Z(s)$ explains input voltage divided via filter components to produce output voltage.

Transfer function describes LCL filter responds different input signal frequencies.

Derivation of LCL Filter Equations

Applying Kirchhoff's voltage law (KVL)

Derive mathematical equations, apply KVL to loop consisting of L_1 , C , and R_{Load} .

$$V_{in} = V_{L1} - V_C - V_{Rload} = 0 \quad (3)$$

- $V_{L1} = L_1 \frac{di}{dt}$ Voltage across L_1 .

- $V_C = C \int idt$ Voltage across C .

- $V_{Rload} = R_{Load} \cdot i$ Voltage across R_{Load} .

This equation applies Kirchhoff's voltage law to loop formed by L_1 , C , and R_{Load} .

It states that sum of voltages across each component in the loop equals input voltage V_{in} , it is starting point for deriving time-domain equations of LCL filter.

Combining voltage expressions, substitute expressions for

V_{L1} , V_C , and V_{Load} into equation (3).

$$V_{in} = L_1 \frac{di}{dt} + \frac{1}{C} \int idt + R_{Load} \cdot i \quad (4)$$

- This equation express input voltage V_{in} regarding current i and derivatives.
- $L_1 \frac{di}{dt}$. Voltage via inductor L_1 depends on current rate of change.
- $\frac{1}{C} \int idt$. Voltage via capacitor C depends on integral of the current.
- $R_{Load} \cdot i$. Voltage across load resistor R_{Load} is proportional to the current.

Transformation of time-dependent function $f(t)$ is denoted as $F(s)$, corresponding expression that facilitates this analysis obtained by apply Laplace transform to both sides of given equation, to analysis simplifying, it converts time-domain equation to s-domain using Laplace transform.

$$\mathcal{L}\{V_{in}\} = L_1 s I(s) + \frac{1}{Cs} I(s) + R_{Load} I(s) \quad (5)$$

The equation converts time-domain equation (4) to s-domain by Laplace transform.

- $\mathcal{L}\{V_{in}\}$ input voltage Laplace transform V_{in} .
- $I(s)$ current Laplace transform $i(t)$.
- $L_1 s I(s)$ s-domain representation of voltage across L_1 .
- $\frac{1}{Cs} I(s)$ s-domain representation of voltage across C .
- $R_{Load} I(s)$ s-domain representation of voltage across R_{Load} .

Determining current $I(s)$ when input voltage $\{V_{in}\}(s)$ is provided, once it obtains $I(s)$, inverse Laplace transform apply to derive correspond time-domain current $i(t)$, this

approach links frequency domain representation to time-domain behaviour, facilitating understanding of circuit dynamics.

The current $I(s)$ solution expressed in terms of input voltage $\{V_{in}\}(s)$. solving for current $I(s)$, rearranging equation (5) to solve for $I(s)$.

$$I(s) = \frac{V_{in}(s)}{L_1 s + \frac{1}{Cs} + R_{Load}} \quad (6)$$

- This equation solves current $I(s)$ in s-domain.
- Denominator $L_1 s + \frac{1}{Cs} + R_{Load}$ explains total impedance of LCL filter.
- The current $I(s)$, input voltage $V_{in}(s)$ divided by total impedance.

Using Laplace transform, it derived output voltage $V_{out}(s)$ via load resistor R_{Load} in LCL filter, this final expression offers improves into various circuit components influence the output in s-domain, output voltage $V_{out}(s)$ corresponds to voltage via the resistor, that expressed as R_{Load} . $I(s)$, this formulation highlights the relationship between load resistor and current flowing through it, for analyzing circuit performance, output voltage $V_{out}(s)$ via load resistor R_{Load} .

$$V_{out}(s) = R_{Load} \cdot \frac{V_{in}(s)}{L_1 s + \frac{1}{Cs} + R_{Load}} \quad (7)$$

- Equation calculates output voltage $V_{out}(s)$ via load resistor R_{Load}
- Output voltage is proportional to the current $I(s)$ flowing through R_{Load}

Transfer function $H(s)$ defines relationship between Laplace transform of output voltage $V_{out}(s)$ and Laplace transform of input voltage $V_{in}(s)$ as a ratio.

This formulation characterizes system's response to varying frequencies within input signal, providing important improves into its behaviours and stability, transfer function summarize system's dynamics in s-domain, making it critical tool in control systems, by explaining ratio of output voltage to input voltage, $H(s)$ illustrates relationship of these two voltages.

Transfer function $H(s)$ is the ratio of $V_{out}(s)$ to $V_{in}(s)$.

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{R_{Load}}{L_1 s + \frac{1}{Cs} + R_{Load}} \quad (8)$$

Transfer function explains frequency response of LCL filter. LCL filter's transfer function explains input voltage $V_{in}(s)$ is transformed into output voltage $V_{out}(s)$.

Transfer function analyses filter's frequency response and stability.

Transfer function $H(s)$ summarize important features of LCL circuit, explain relationship between output voltage $V_{out}(s)$ and input voltage $V_{in}(s)$ as influenced by various circuit components, transfer function reflects how these components interact within LCL filter, and it is expressed as follows.

$$H_{LCL}(s) = \frac{R_{Load}}{L_1 s + \frac{1}{Cs} + R_{Load}} \quad (9)$$

Transfer function of LCL filter in grid-tied PV-microinverter help design and tune output filter for required performance and stability, hybrid filter transfer function is described in detail, considering LCL and PD filter, transfer function for PD filter gives by.

$$H_{PD}(s) = \frac{R_d}{(L_d s + \frac{1}{C_d s}) + R_d} \quad (10)$$

Two systems are connected in parallel, transfer function derived using concept of parallel impedance, for parallel systems, total output is sum of outputs from individual systems at same input is applied, total transfer function

$H_{hybrid}(s)$ is obtained by combining transfer functions of LCL and PD filters.

$$H_{hybrid}(s) = \frac{H_{LCL}(s) \cdot H_{PD}(s)}{H_{LCL}(s) + H_{PD}(s)} \quad (11)$$

Total transfer function, $H_{hybrid}(s)$, is obtained by combining transfer functions of LCL filter and PD filter connected in parallel, that compact formula provides combined transfer function of LCL and PD filters, numerator explains combined gain from both filters, and denominator implements combined impedance of both filters' frequency-dependent and resistive elements, substitute the expressions for $H_{LCL}(s)$ and $H_{PD}(s)$ into the equation.

$$H_{hybrid}(s) = \frac{\frac{R_{Load}}{L_1 s + \frac{1}{C_s} + R_{Load}} \cdot \frac{R_d}{(L_d s + \frac{1}{C_d s}) + R_d}}{\frac{R_{Load}}{L_1 s + \frac{1}{C_s} + R_{Load}} + \frac{R_d}{(L_d s + \frac{1}{C_d s}) + R_d}} \quad (12)$$

To simplify expression for hybrid transfer function of LCL and PD filters connected in parallel, start by identifying each component in transfer function equations, these transfer functions are substituted with total hybrid transfer function formula, focus on the denominator, that include sum of two complex fractional expressions, the result shows hybrid transfer function in the form that indicates how the components of LCL and PD filters contribute to the system's behaviour, simplify expression by finding denominator and combining terms.

$$H_{hybrid}(s) = \frac{R_{Load} R_d}{(L_1 s + \frac{1}{C_s} + R_{Load})((L_d s + \frac{1}{C_d s}) + R_d) + R_{Load} R_d} \quad (13)$$

Hybrid transfer function $H_{hybrid}(s)$ is derived by combining individual transfer functions of LCL and PD filters when connected in parallel, total transfer function is obtained by multiplying individual transfer functions and dividing by their sum. Final transfer function.

$$H_{hybrid}(s) = \frac{R_{Load} R_d}{L_1 R_d s^2 (L_1 C_d + L_1 R_{Load} + R_d C) s + (C R_d + R_{Load} R_d + L_1)} \quad (14)$$

Detailed transfer function for hybrid filter, considering both LCL and PD filters, it used to analyze and design hybrid filter in two-stage grid-tied PV-microinverters.

Reduction of grid current THD using PD filter, design, and implementation of hybrid filter, must be relevant IEEE and IEC standards and guidelines.

Let I_{grid} (t) grid current, and I_{grid_fund} (t) its fundamental frequency component, THD is defined as ratio of harmonic content to RMS of the fundamental component.

$$THD = \frac{\sqrt{I_{grid}^2 - I_{grid_fund}^2}}{I_{grid_fund}} \quad (15)$$

Hybrid filter influences dynamic response of grid current, rapid transients in the inverter current are shaped by filter according to its design parameters, notably cutoff frequencies, determining extent and speed of current modulation.

Let consider impact of hybrid filter transfer function

$H_{hybrid}(s)$ on grid current.

$$I_{grid}(t) = H_{hybrid}(j\omega) \cdot I_{inverter}(t) \quad (16)$$

Filter's magnitude response determines amplitude scaling and harmonic reduction at specific frequency, improve power quality system efficiency, simplify focus on frequency domain (ω).

$$I_{grid_fund}(\omega) = |H_{hybrid}(j\omega)| \cdot I_{inverter_fund}(\omega) \quad (17)$$

$I_{inverter_fund}(\omega)$ shows fundamental frequency component of inverter current, it assumes that hybrid filter attenuates harmonic components, $H_{hybrid}(j\omega) = 0$ for ($\omega \neq \omega_{fund}$), it expresses THD reduction.

$$THD_{filter} = \frac{\sqrt{I_{grid_filter}^2 - I_{grid_fund_filter}^2}}{I_{grid_fund_filter}} < THD_{inverter} \quad (18)$$

THD_{filter} grid current THD through hybrid filter, $THD_{inverter}$ inverter current THD, I_{grid_filter} grid current through hybrid filter, $I_{grid_fund_filter}$ main frequency of grid current through hybrid filter.

4.0 RESULTS AND DISCUSSION

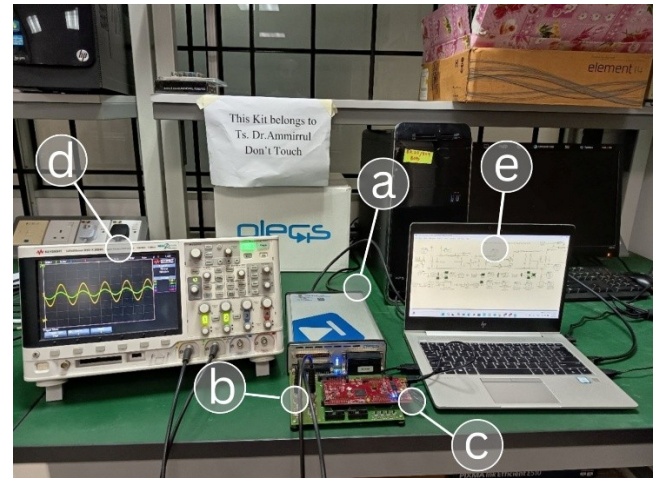


Figure 4 HIL Experimental Setup: (a) PLECS RT-Box (b) RT-Box Launch Pad Interface (c) Texas Instruments Launch TMS320F28379D (d) Keysight Oscilloscope (e) Computer to program RT-Box and Microcontroller

Figure (4) shows HIL experimental setup designed to analyze performance of two-stage grid-tied PV-microinverter at various filter topologies aimed at reducing grid current total harmonic distortion THD, the system consists of PLECS hardware instrument, that works as real-time simulation platform for PV-microinverter's power stage and control algorithms, interfaced with computer running PLECS

software to model converter and evaluate different passive filters.

The input signals from computer are processed in real time by PLECS hardware, emulating dynamic behaviour of PV system at varying operational conditions, while output waveforms including grid voltage and current, inverter current, and filtered output current are captured and displayed on Keysight oscilloscope for accurate THD measurement and waveform analysis, this HIL-based verification approach ensures accurate testing of filter topologies as physical implementation, bridging gap between simulation and practical implement and providing important improves into harmonic mitigation strategies for grid-connected PV systems.

4.1 HIL-PLECS RESULTS, LCL FILTER AT SS.

Figure (5) shows grid voltage measured at 325V during steady-state operation, as recorded using Keysight oscilloscope within HIL-PLECS setup and equipped with all filter topologies LCL, H-B filter and PD filter, this measurement indicates stable and constant voltage level at all filter topologies, reflecting effective performance of grid-connected system at operation.

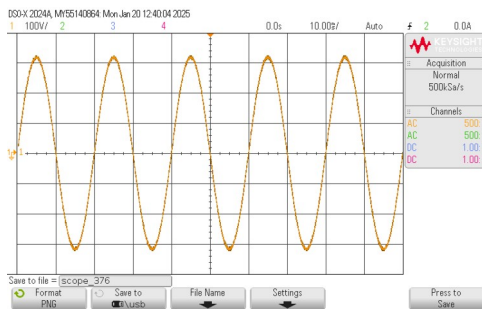


Figure 5 Grid voltage, LCL filter at steady state

Figure (6) presents experimental results of grid current measured at 2.01A at steady-state of PV-microinverter, measurements obtained using Keysight oscilloscope integrated with HIL-PLECS setup, implementing LCL filter, THD of grid current from experimental results of two-stage PV-microinverter using LCL filter at steady-state, the measured THD is 11.33%, higher than acceptable threshold, to comply with IEEE standards power quality, this THD must be reduced to below 5%, this confirms importance of implement filtering strategies, to reduce harmonic distortion and enhance performance of grid-tied system.

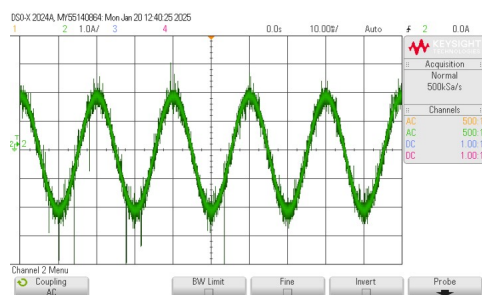


Figure 6 Grid current, LCL filter at steady state

in Figure (7) detects total harmonic distortion THD of 11.33%, which exceeds 5% limit in IEEE 519-2014 and IEC 6000 standards power quality, this higher THD indicates substantial harmonic in output power, leading to grid compatibility issues and reduced system efficiency, the excessive distortion indicates that, LCL filter design requires optimization to better harmonics attenuation, possible new filter topology implementation of passive damping techniques to obtain compliant power quality standards.

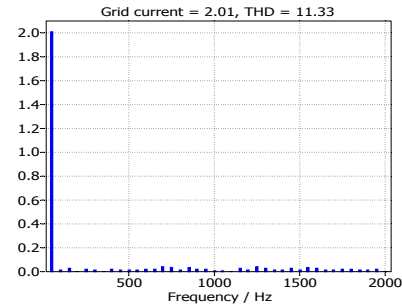


Figure 7 Grid current THD, LCL filter at steady state

4.2 HIL-PLECS RESULTS, LCL+H-B FILTER AT SS.

Figure (8) presents experiment results of grid current measured at 4.15A during steady-state of two-stage grid-tied PV-microinverter, measurements obtained using Keysight oscilloscope integrated with HIL-PLECS setup, implementing H-B hybrid filter.

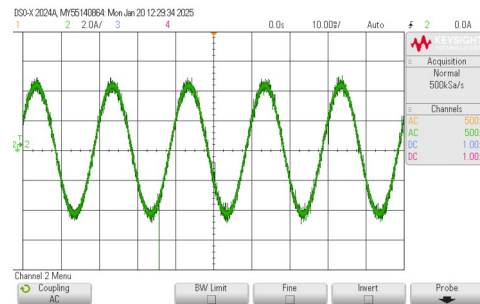


Figure 8 Grid current, LCL+H-B filter at steady state

The 4.15A current output confirms system's ability to maintain stable and substantial power injection into grid system at steady state, close relation between experiment and simulation results provides strong conformation for reliability of implemented control strategy and hardware configuration.

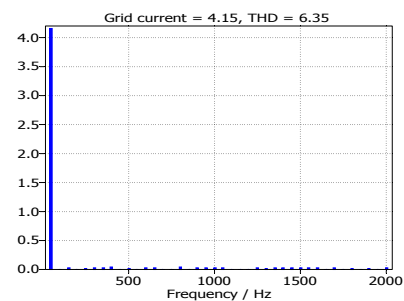


Figure 9 Grid current THD, LCL+H-B filter at steady state

THD is 6.15%, higher than acceptable value of 5% established by IEEE and IEC power quality standards, this higher THD lead to various issues, as increased heating in electrical devices and reduced system efficiency, to ensure compliance with power quality standards, and optimize performance of PV-microinverter, it is important to implement topology reducing THD to below 5%.

4.2 HIL-PLECS RESULTS, LCL+PD FILTER AT SS.

Figure (10) shows steady-state performance of two-stage PV-microinverter system, showing measured grid current of 4.23A at implementing the PD hybrid filter, experiment measurements, obtained through Keysight oscilloscope integrated with HIL-PLECS setup results.

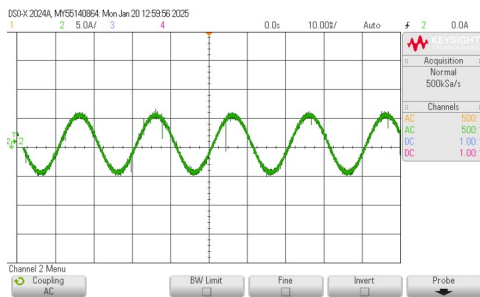


Figure10 Grid current, LCL+PD filter at steady state

The measured THD is **3.75%**, it's an excellent result that is below the 5% threshold established by IEEE standards power quality, this low THD indicates that grid current waveform is clean, harmonics reduction by PD filter complies with regulatory standards.

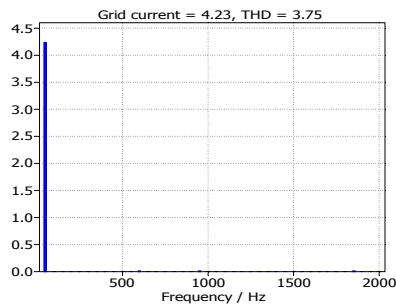


Figure 11 Grid current THD, LCL+PD filter at steady state

PD filter's reducing THD of grid current impacts system efficiency through two mechanisms, first, improved waveform quality reduces harmonic-related losses in both microinverter and grid-tied equipment, second, lower THD minimizes reactive power component, increasing system's effective power factor and real power transfer ability as shows in Table. 1.

Table 1 Model test specifications and test conditions

Filters	I_{out}	V_{out}	P_{out}	THD	Efficiency
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LCL Filter	2.01A	325V	653.3W	11.33%	97%
LCL+H-B Filter	4.15A	325V	1.349W	6.35%	91%
LCL+PD Filter	4.23A	325V	1.375W	3.75%	94%

5.0 CONCLUSION

This paper examined the efficiency of three different filter methodologies implement to two-stage grid-tied PV-microinverter, focusing on their performance to decrease grid current THD, analysis compared LCL filter, H-B filter and PD filter, Each filtering approach evaluated based on its ability to meet harmonic distortion standards set forth by IEEE and IEC, While wide used in grid-tied applications due to its simplicity and effective performance, LCL filter showed THD ranging from 11% to 12%, this THD exceeds acceptable limits set by IEEE and IEC standard for grid-tied systems, LCL filter offers fundamental baseline for performance, it represents need for more developed filter solutions for compliance and improved power quality.

H-B filter exhibited improved performance, obtaining THD reduction between 6% and 7%, this methodology combines strengths of both LCL and H-B filters, result more mitigation of harmonics and maintain system stability, use of H-B filter contributes to more comprehensive harmonic attenuation strategy, THD were lower with this approach, they still did not meet stringent requirements imposed by IEEE and IEC standards.

Best results observed with PD filter combination, that reduced THD between 3% and 4%, this method provides higher harmonic reduction and enhances system's stability by dampening oscillations, reduction in THD produced by PD filter it as optimal solution for ensuring compliance with IEEE and IEC standards, meeting these standards is important to enhancing power quality, lower THD is important for improving system efficiency and reliability and integration of renewable energy sources.

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